All-pairs shortest paths: exercises

25-E01 The second step of two different DPs

Use1→2:4,1→3:7,2→3:−2,2→4:6,3→4:3,4→2:1. Give every row of $L^{(2)}$ and $D^{(2)}$, identify at least two differing entries, and explain their state restrictions. Calculate all four candidates for $L^{(2)}[1,4]$, then find $L^{(3)}[1,4]$.

25-E02 Negative diagonals and locally affected pairs

Use1→2:2,2→3:−3,3→2:1,3→4:4, and isolated5. Classify all 25 distances as finite,∞, or−∞. Explain cycle entry and exit for δ(1,1),δ(1,4),δ(4,1),δ(5,5). If final Floyd has D[2,2] <0, may that temporary finite value be returned as δ(2,2)?

25-E03 Johnson's cost model

For n=1000, m=5000, give leading expressions for Floyd, all-zero-potential Bellman–Ford, repeated binary Dijkstra, and repeated Fibonacci Dijkstra. State the full-output space lower bound. For m=0, an explicit super-source adds n edges. How many edges does Bellman–Ford process, and why is nm=0 an invalid bound for that implementation?

25-E04 Potential and distance restoration

On25-E01, synchronously relax all-zero h to stability. List h and every $\hat{w}$. Trace Dijkstra from 1, including candidate changes 9→4 for 3 and 10→5 for 4. Restore all original distances from 1, then restore3→2 and 4→3 to check the signs.

25-E05 Both directions of nonnegative reweighting

Prove that a finite h with $w(u, v)+h(u)-h(v) \geq 0$ on every edge exists if and only if the graph has no negative cycle. Give a numerical example with routes of different edge counts where adding one common constant per edge changes the shortest route. Why does potential reweighting avoid that problem?

25-E06 Predecessors with a zero cycle

Use1→2:0,2→1:0,1→3:1,2→3:1. For source 1, list optimal(cost, hops) labels, choose parents tight in both coordinates, and reconstruct a route to 3. Give cyclic parents that cost tightness alone permits. With only edge 1→2:1, explain the problem with updating Π[i, j]=Π[k, j] on equal costs when k=j.

All-pairs shortest paths: exercises (continued)

25-E07 Associativity and loop order

Expand the min-plus product of three n×n matrices and prove associativity. Explain why an n=6 graph without negative cycles permits budgets1→2→4→8. On unit chain1→2→4→3, show entry by entry why one i, j, k pass misses1→3. How does this differ from the proof for in-place k, i, j?

> Complete the derivation before consulting the separate solutions PDF.

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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https://github.com/xinhi6716/computer-science-self-study

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